F FWallSim · Firewall Course

CLASS 0

Course plan

19 classes, about 6 hours, in the order to take them

Beginner | ~335 minutes | FWallSim v1.0.0 (FortiOS 7.4 style)

YOU WILL LEARN

- Which class to take next
- What each class covers
- Which PDF file belongs to each class

BEFORE YOU START

Start with Class 1 and go in order. Each class builds on the one before it.

How every class works

1. Read the goal and look at the lab network diagram.
2. In FWallSim click **Labs**, find the class title and click **Start**.
3. Do the numbered tasks. The Lab Guide on the right ticks each task green the moment it works.
4. Use the screenshots when you are not sure where to click.
5. In the troubleshooting classes, try the hints first and look at the solution last.

Part 1: Beginner, the basics

Class	Title	Topics	Time	PDF
1	First login and system basics	GUI, System settings, local-in	~10 min	`FWallSim-Class-01-l1-basics.pdf`
2	Your first firewall policy: LAN to Internet	Firewall policy, NAT, implicit deny	~15 min	`FWallSim-Class-02-l2-first-policy.pdf`
3	Logs: seeing what the firewall did	Log Allowed Traffic, implicit deny logging, Forward Traffic log	~10 min	`FWallSim-Class-03-l3-logging.pdf`
4	Objects: addresses and service groups	Address objects, service groups, least privilege	~15 min	`FWallSim-Class-04-l4-objects.pdf`
5	Publish a web server with a Virtual IP	Destination NAT (VIP), inbound policy	~20 min	`FWallSim-Class-05-l5-vip.pdf`

Part 2: Intermediate, running the firewall

Class	Title	Topics	Time	PDF
6	Rule order matters	First-match policy lookup, policy order	~15 min	`FWallSim-Class-06-l6-order.pdf`
7	Change the LAN subnet	Interfaces, DHCP server, address objects	~20 min	`FWallSim-Class-07-l7-renumber.pdf`
8	Harden administrative access	Admin profiles, trusted hosts, local-in policy	~20 min	`FWallSim-Class-08-l8-harden.pdf`
9	Identity-based access with LDAP	LDAP server, user groups, identity-based policy	~25 min	`FWallSim-Class-09-l9-ldap.pdf`

Part 3: Advanced, VPN and routing

Class	Title	Topics	Time	PDF
10	Site-to-site IPsec VPN	IPsec phase 1/2, routes, VPN policies	~30 min	`FWallSim-Class-10-l10-ipsec.pdf`
11	SSL-VPN for remote workers	SSL-VPN settings, portals, ssl.root policy, split tunneling	~25 min	`FWallSim-Class-11-l11-sslvpn.pdf`
12	SD-WAN with two ISPs	SD-WAN members, performance SLA, failover	~30 min	`FWallSim-Class-12-l12-sdwan.pdf`
13	Dynamic routing: OSPF over VPN and BGP	OSPF neighbors, BGP peering, administrative distance	~30 min	`FWallSim-Class-13-l13-dynamic.pdf`

Part 4: Troubleshooting challenges

Class	Title	Topics	Time	PDF
14	Nobody can browse the internet	Troubleshooting: NAT	~10 min	`FWallSim-Class-14-c1-no-internet.pdf`
15	Pings work, websites do not	Troubleshooting: DNS	~10 min	`FWallSim-Class-15-c2-dns.pdf`
16	Our website is down	Troubleshooting: VIP	~10 min	`FWallSim-Class-16-c3-website.pdf`
17	PC2 has no network	Troubleshooting: DHCP	~10 min	`FWallSim-Class-17-c4-dhcp.pdf`
18	The branch office is cut off	Troubleshooting: IPsec	~15 min	`FWallSim-Class-18-c5-branch.pdf`
19	The backup line never takes over	Troubleshooting: SD-WAN health checks	~15 min	`FWallSim-Class-19-c6-failover.pdf`

Next: Class 1: First login and system basics